

SymPy Bug Card

Bug 50: Wrong negative-branch antiderivative for $\operatorname{asec}(x)$

Task. Integrate inverse secant and check the returned primitive on the negative real branch.

$$F(x) = \operatorname{integrate}(\operatorname{asec}(x), x), \quad x = -2.$$

SymPy's answer. SymPy returns a primitive whose derivative has

$$F'(-2) - \operatorname{asec}(-2) = -\frac{2\sqrt{3}}{3}.$$

Correct answer. The derivative residual should be 0. On $x < -1$, one valid real branch is

$$F(x) = x \operatorname{asec}(x) - \operatorname{acosh}(|x|) + C.$$

Explanation. Integration by parts leaves the subintegrand $1/(x\sqrt{1-1/x^2}) = -1/\sqrt{x^2-1}$ on $x < -1$, whose primitive is $\operatorname{acosh}(|x|)$. SymPy uses the opposite real sign there, so the final antiderivative differentiates to $\operatorname{asec}(x) - 2/\sqrt{x^2-1}$, giving the exact residual above at $x = -2$.

Likely root cause. The diagnosis traces the sign error to the Meijer-G/`hyperexpand` path in `HyperRep_asin1`: after rewriting with $z = x^2$, the expansion introduces $x \operatorname{acosh}(|x|)/|x|$ instead of preserving the negative-branch primitive $\operatorname{acosh}(|x|)$. The asec antiderivative itself was added by PR #19993 (closing issues #14329 and #19983), and the same wrong-real-branch Meijer/`acosh` mechanism is reported in issue #10453; the specific failure here—a *nonzero* derivative residual on the smooth branch $x < -1$ —is the unreported instance of that known family.

Artifact (GitHub).

[bug-report/bug-050-wrong-negative-branch-antiderivative-for-asec-x/](#)